

Fraud Detection Suite: Real-Time Risk & Intelligence

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Target Stakeholders: Fraud Operations, Risk Management, & Chief Risk Officers (CRO)

Executive Summary

The **Fraud Detection Suite** is an executive-grade decisioning platform designed to mitigate financial leakage through real-time predictive scoring and geospatial transaction intelligence. By transforming raw transaction logs into actionable risk signals, the suite enables institutions to move from reactive mitigation to proactive prevention, directly supporting **NACHA 2026 Fraud Monitoring** requirements.

Strategic Business Value:

- **Operational Resilience:** Automates initial triage for high-velocity environments, targeting a **sub-100ms alert generation** benchmark to stop fraud at the point of sale.
- **Precision-Based Triage:** Leverages a Random Forest Classifier to reduce "over-blocking," aiming for a **false positive rate below 15%** to protect legitimate customer revenue.
- **Regulatory Readiness:** Built to meet Phase 1 (March 2026) and Phase 2 (June 2026) Nacha rule changes requiring risk-based processes to identify atypical transaction patterns.

Strategic Modules & Intelligence

The portal features a "Human-in-the-Loop" architecture designed for high-fidelity investigation and rapid case resolution.

Predictive Risk Engine

- **Goal:** Score every transaction for potential fraud using behavioral and financial telemetry.
- **Logic:** Utilizes a **Random Forest Classifier** trained on volume, merchant category, and "distance from home" outliers (calculated via the Haversine formula).
- **Versioned Health:** Tracks model drift over time to ensure detection baseline remains resilient against evolving fraud typologies.

 Geospatial Intelligence (GIS)

- **Goal:** Visualize the "where and when" of fraud to identify regional "hot zones."
- **Visualizations:** Integrates **PyDeck** with `ArcLayer` to map the physical displacement between a user's home base and the transaction origin.
- **Hot Zone Mapping:** Uses `HeatmapLayers` to identify geographic clusters of fraud, enabling regional-level policy adjustments.

 Case Management & Merchant Intel

- **Goal:** Facilitate deep-dive audits of suspicious accounts and merchants.
- **Analytics:** Benchmarks individual transactions against historical merchant averages and monitors "spending velocity" to differentiate genuine surges from fraudulent spikes.

 Technical Implementation

This project demonstrates proficiency in integrating complex machine learning pipelines with high-fidelity, interactive frontends.

Layer	Technology Stack
Frontend	Streamlit (Session state management & multi-tab navigation)
Intelligence	Scikit-Learn (Random Forest), Pandas , NumPy
Visualization	PyDeck (Mapbox integration), Plotly Express (Risk distribution)

Strategic KPI Glossary

Metrics calculated within the suite are aligned with **2026 Executive Fraud Reporting** standards:

- **Fraud-to-Sales Ratio:** High-level financial exposure; benchmarked against an industry standard of **less than 1%**.
- **Precision Rate:** The ratio of fraudulent transactions to total declines, ensuring minimal customer friction.
- **Mean Time to Detect (MTTD):** Measures the agility of the system in identifying and flagging new fraud trends as they emerge.

Installation & Access

To launch the Fraud Detection Portal locally:

1. **Clone the repository:** `git clone https://github.com/your-username/fraud-detection-suite.git`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Launch the application:** `streamlit run app.py`

Admin Gateway Credentials:

- **Username:** admin
- **Password:** Risk2026

Repository Strategy

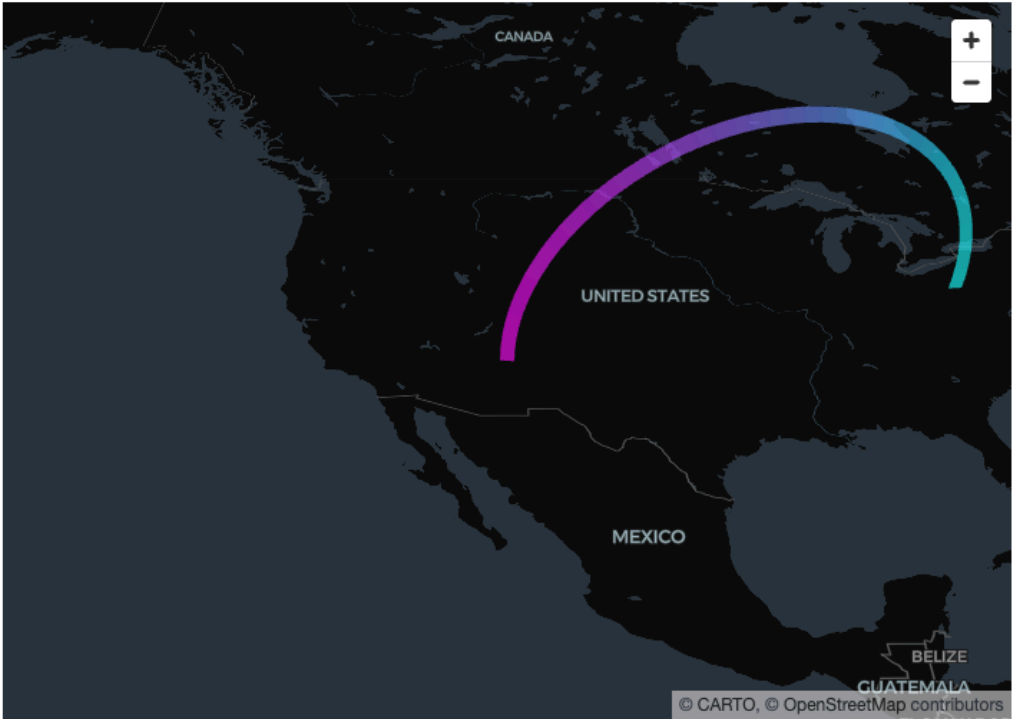
This repository serves as a **Narrative of Authority**, shifting the focus from simple monitoring to institutional resilience.

Operational Focus (Old)	Strategic Focus (New)
Lists "Geospatial Map" as a feature.	Frames it as "Geospatial Intelligence" for Day-0 detection.

Operational Focus (Old)	Strategic Focus (New)
Describes ML as "Random Forest."	Frames it as "Precision-Based Triage" to protect ROTCE.

 Screenshots

Case ID: 0 | Los Angeles, CA

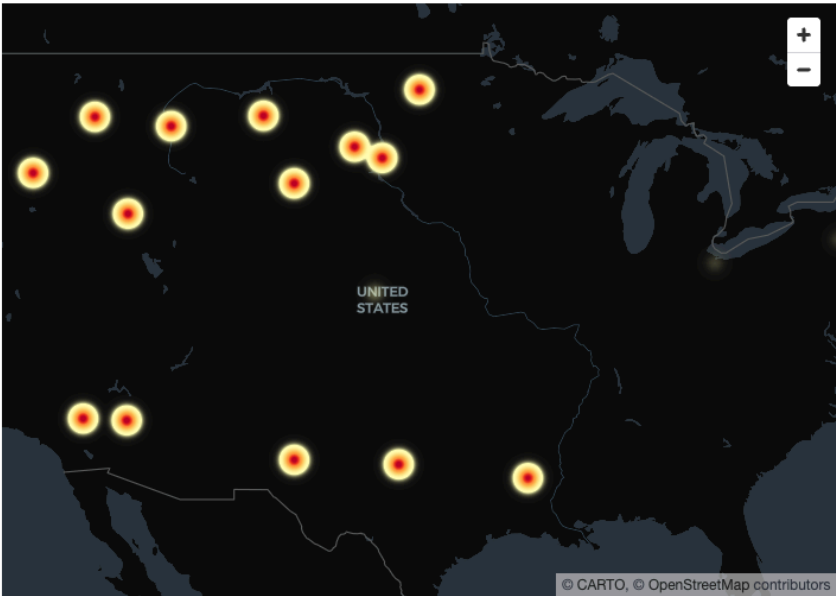


id	timestamp	merchant	Location	amount	risk_score	priority
165	2026-04-08 17:28:07.581062	Retail	Chicago, IL	265.84	0	 Low
388	2026-04-08 15:14:07.581908	Gas	Miami, FL	658.37	0	 Low
699	2026-04-08 13:58:07.583065	Grocery	Seattle, WA	1119.88	100	 High
644	2026-04-08 13:35:07.582865	Retail	Seattle, WA	1078.71	100	 High
906	2026-04-08 13:05:07.583838	Grocery	Miami, FL	956.02	0	 Low
440	2026-04-08 10:42:07.582099	Grocery	Los Angeles, CA	889.18	100	 High

Geographic Risk Density

📍 **How to read this map:** Heat clusters represent cumulative risk. Bright red areas indicate high-frequency fraud attempts occurring within the selected hour.

Filter by Hour (24h Format):



id	merchant	Location	amount	risk_score	priority
0	Retail	Los Angeles, CA	882.21	100	 High



id	merchant	Location	amount	risk_score	priority
0	Retail	Los Angeles, CA	882.21	100	High
1	Retail	Miami, FL	769.13	98	High
2	Grocery	Seattle, WA	397.56	3	Low
3	Digital	Miami, FL	290.58	0	Low
4	Grocery	Miami, FL	949.72	100	High
5	Gas	Miami, FL	575.34	0	Low
6	Grocery	Seattle, WA	516.32	1	Low
7	Retail	Los Angeles, CA	538.50	0	Low
8	Digital	Miami, FL	531.94	0	Low
9	Grocery	Seattle, WA	478.40	0	Low

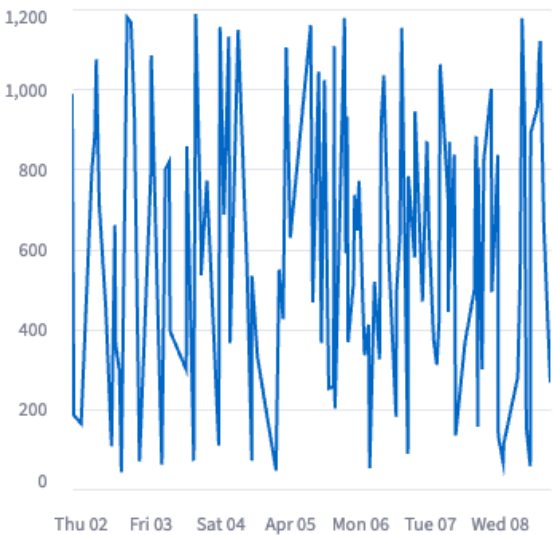
Fraud Detection Suite v1

-  Case Analysis
-  Global Heatmap
-  Merchant Intel
-  Model Health

Merchant Intelligence & Behavior Analysis

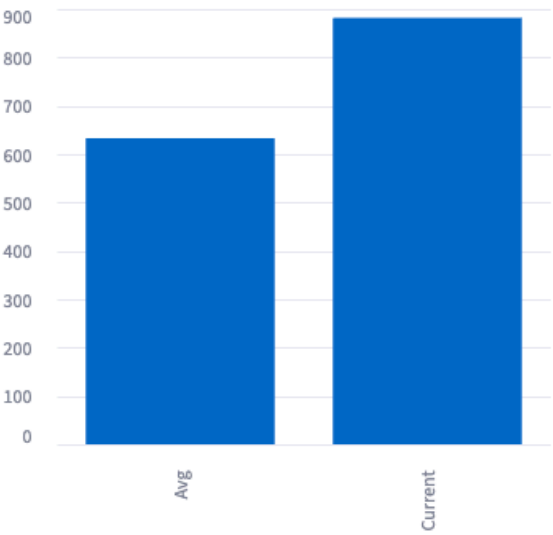
This tab provides a deep dive into individual user spending patterns and category-wide risk trends.

User 1005 Spending Velocity



Velocity Analysis: Sudden vertical spikes signal outlier behaviors deviating from historical norms.

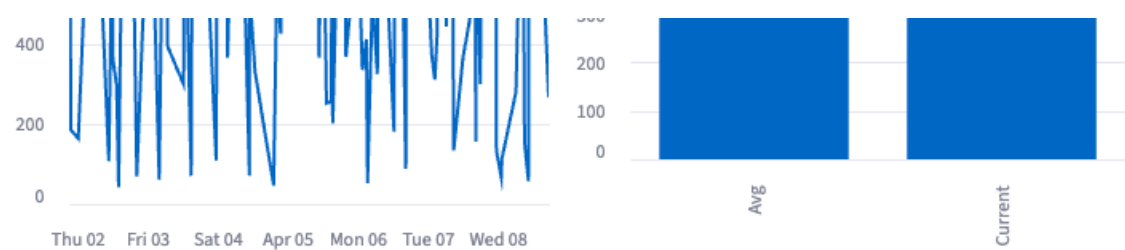
Merchant Context: Retail



Peer Benchmarking: Compares current amount against historical merchant average.

 Rolling Merchant Risk Velocity (24h)

Rolling Merchant Risk Velocity (24h)



Velocity Analysis: Sudden vertical spikes signal outlier behaviors deviating from historical norms.

Peer Benchmarking: Compares current amount against historical merchant average.

 Rolling Merchant Risk Velocity (24h)

 Filter Velocity by Geography:

Northeast 

Merchant	Velocity	Current Avg Risk
Gas	0.26	0.6
Digital	-0.06	0.2
Grocery	-0.1	0
Retail	-0.16	0

 **Note:** Trends for **Northeast** region. Velocity > 5.0 indicates emerging risk clusters.



Fraud Detection Suite v1

 Case Analysis

 Global Heatmap

 Merchant Intel

 **Model Health**

System Performance & Auditor Oversight

Current Baseline Risk

41.31%

Model Version

v1

↑ Active

 Retrain Model & Refresh Data



Version Comparison History

Version	Baseline Risk	Timestamp
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41.31%

v1

↑ Active

 Retrain Model & Refresh Data

 Version Comparison History

Version	Baseline Risk	Timestamp
1	41.31%	17:50:07

Feature Importance

